

Review of Related Literature and Studies

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This chapter reviews the literature that grounds the proposed benchmarking of multitask optimization (MTO) on Taglish disaster tweets. The relevant work spans three areas: crisis informatics, MTO, and code-switched text processing. No single area establishes whether specialized balancing benefits triage on code-switched text, so the review covers the three areas in sequence. It arranges prior work by computational subconcept, algorithmic paradigm, and empirical bottleneck rather than by geographical origin.

Section 2.1 reviews crisis informatics corpora, joint-modeling evidence for entity, intent, and urgency tasks, cross-event and cross-lingual generalization, and the computational, statistical, and methodological standards the benchmark adopts. Section 2.2 then turns to MTO. It analyzes the gradient-surgery and dynamic loss-weighting families, the optimizer confound, and the controlled comparisons questioning whether specialized balancing outperforms tuned static linear scalarization. Section 2.3 narrows to the target domain, covering code-switching challenges and measurement, multitask learning on code-switched text, multilingual encoders, and Philippine language resources, including local disaster response studies. Section 2.4 consolidates the key studies in the Algorithmic Literature Taxonomy Matrix, which positions them against the proposed research. Section 2.5 closes with a three-step synthesis that derives the research gap and maps the study’s contributions onto it.

Database searches conducted as of 2026 on MTO, crisis informatics, and code-switching identified the sources. These searches drew primarily on peer-reviewed

conference and journal publications indexed in the ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, and the Association for Computational Linguistics (ACL) Anthology. Within this set, the crisis informatics and code-switching sources that established the evaluation domain date mostly from the last five years. The review retained older studies only if they defined a benchmarked algorithm, a candidate encoder, a methodological and statistical protocol, or a foundational corpus or metric.

2.1 Joint Multitask Modeling and Computational Efficiency in Crisis Informatics

Section 2.1 traces the crisis informatics arc from annotated corpora through joint triage models to cross-event generalization. It then sets the computational, statistical, and methodological standards the benchmark adopts.

2.1.1 Evolution of Crisis Informatics Corpora and Models

Both the corpus track and the model track begin with a demonstration that disaster tweets can be annotated and classified at scale. Imran et al. (2016) collected messages during 19 crises between 2013 and 2015 and released annotations for approximately 50,000 of them. The authors also published word embeddings trained on 52 million crisis-related tweets. The annotation schemes incorporated input from formal response agencies such as the United Nations Office for the Coordination of Humanitarian Affairs. The resulting nine categories included “caution and advice,” “displaced people and evacuations,” and “infrastructure and utilities damage.” Naive Bayes, random forest, and support vector machine classifiers reached an area under the curve (AUC) of at least .80 for most categories, versus .50 for random classification. The exception was the smallest class, “missing, trapped, or found people.” The curation pipeline also flagged

out-of-vocabulary terms such as slang, abbreviations, and misspellings, a noise thread that Section 2.3 develops.

Corpus supply then kept growing, but dispersed label schemes left results impossible to compare. Alam, Sajjad, et al. (2021) documented that gap and responded with CrisisBench, consolidating eight human-annotated datasets into two corpora of 166,098 and 141,533 tweets. The curation pipeline eliminated three forms of duplication, pairing exact string matching with cosine similarity filtering to prune retweets and copied posts. That filtering kept duplicate messages out of both training and test splits.

Corpus construction then scaled up under explicit agreement control. Alam, Qazi, et al. (2021) released the Human-Annotated Disaster Incidents Data (HumAID), sampling tweets from a pool of about 24 million messages across 19 disaster events between 2016 and 2019. Crowd workers read 77,196 sampled tweets under three judgments per tweet, and a label was kept when two of the three agreed, a two-thirds (66%) agreement threshold. Combining the events for cross-event experiments triggered a further near-duplicate removal that left 76,466 tweets.

On these standardized benchmarks, fine-tuned encoders displaced the classical classifiers. CrisisBench’s consolidated humanitarian task set the pattern: A fine-tuned RoBERTa model reached .872 on the weighted F1-measure (F1), against .829 for the strongest convolutional-network baseline. A fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model trailed RoBERTa by only .012 weighted F1 on the same task. Evaluations on HumAID showed the same ordering across events. RoBERTa reached .781 average weighted F1, whereas the support vector machine, random forest, and fastText classifiers stayed at or below .712. Two cautions qualify these

numbers. Weighted F1 averages per-class scores with class-frequency weights, so majority classes dominate the total; the benchmark adopted it precisely to factor in class imbalance. The two benchmarks differ in corpora, label sets, and splits, so their numbers support separate conclusions rather than one ranking.

Pre-training itself then moved into the crisis domain. Lamsal et al. (2024) trained CrisisTransformers on more than 15 billion tokens drawn from over 30 crisis events and evaluated the models on 18 crisis-specific public datasets. The domain-adapted models outperformed strong baselines across all of those datasets on classification, and the best sentence encoder improved the state of the art by 17.43%. The pre-training run, however, occupied six A100 GPUs for two months, a cost profile that only a cluster supports.

The current frontier couples triage with generative large language models, and it inherits the cluster habit. Yin et al. (2026) instruction fine-tuned a large language model for multilabel disaster classification and reported 63.8% overall accuracy against 4.27% for zero-shot prompted baselines, a nearly fifteenfold margin. LoRA tuning used a 3-GPU A40 node, and full-parameter tuning ran on a 16-GPU A100 node. Zguir et al. (2025) extended triage toward actionable requests and offers through a hierarchical taxonomy and query-specific few-shot learning, evaluating on 300 manually annotated tweets alongside 1,346 synthetic examples. Results of this generation therefore rest on thin evaluation and on hardware beyond a local workstation.

Across both tracks, three absences mark the space the proposed benchmark enters. First, every corpus in the line stopped at document-level labels; none paired token-level entity spans with concurrent sequence-level intent and urgency labels. Second, the systems behind the strongest results trained on clusters or called cloud services. Third, the

corpus benchmarks reported accuracy and F1 but no inference latency or memory. These three absences, not any single result, define the gap, and Section 2.5 returns to them.

2.1.2 Joint Modeling of Token-Level and Sequence-Level Triage Tasks

The joint-modeling question predates crisis informatics by two decades. Caruana (1997) defined multitask learning as inductive transfer that improves generalization by using the domain information in the training signals of related tasks. Models learn tasks in parallel over a shared representation, so what the model learns for one task helps the others. Baxter (2000) supplied the theory: A hypothesis space that performs well on sufficiently many related tasks will also perform well on novel tasks from the same environment.

The nearest structural precedent comes from spoken-language understanding. Weld et al. (2022) surveyed the joint models behind conversational agents and chatbots. The standard approach trains models for intent detection over the whole utterance together with slot filling over its individual tokens. This pairing exhibits the structural configuration that the proposed triage formulation requires: one token-level task alongside sequence-level tasks.

Prior crisis informatics studies applied joint modeling only partially. Wang et al. (2021) fine-tuned one BERT model on the TREC Incident Streams (TREC-IS) crisis corpus to predict information types and priority levels together. The loss weight remained fixed at .50, weighting both task objectives equally. Seeberger and Riedhammer (2022) combined entity-masked language modeling with hierarchical multilabel classification on the same TREC-IS corpus under its 2020B split. The combined design gained up to 10 percentage points of F1 on actionable information types, and entity masking reduced

overfitting to in-domain events.

Prior crisis studies have not evaluated this concurrent pairing. No reviewed study evaluated token-level entity extraction together with intent and urgency classification on the same disaster tweets, and none examined the optimization strategies such architectures demand. Whether specialized balancing resolves that shared loss is the question Section 2.2 takes up.

2.1.3 Cross-Event and Cross-Lingual Generalization

Evaluation practice moved toward held-out events in the same years. Ray Chowdhury et al. (2020) trained multilingual BERT with Manifold Mixup and tested on four datasets, three of which came from disasters absent from the training data. On the English-only evaluation, the model reached an F1 of 79.36% on the Philippines Flood event. Zero-shot transfers scored 81.33% on French, 75.44% on Italian, and 85.26% on Spanish. Per-class evaluation on the multilingual mixed set showed the lowest score on the caution and advice class, at 70.3% F1.

Cross-lingual evaluations also quantified the performance drop incurred during low-resource transfer. Sarioglu Kayi et al. (2020) trained urgency classifiers on English crisis tweets and released evaluation sets for Sinhala and Odia. The ensemble classifier scored 76.5% F1 on English but dropped to 63.5% on Sinhala and 62.6% on Odia when transferred without target-language training. The authors traced part of the noise to annotation because general status updates and critical help requests both carried the urgent label.

Recent work treated generalization failure itself as a modeling problem. Seeberger

et al. (2025) showed that crisis classifiers relied on spurious event-specific cues such as locations and hashtags. The authors proposed a causal debiasing framework with domain-specific experts. Across three disaster classification tasks under temporal splits to unseen events, the approach outperformed the next-best baseline by 1.1 to 1.9 percentage points and significantly improved the BERT baseline. This line of evidence justifies the leave-one-event-out (LOEO) evaluation of the proposed benchmark, which holds out one typhoon at a time.

2.1.4 Complexity Analysis, Statistical Rigor, and Design Science Methodology

The benchmark’s complexity axis rests on a structural property of shared encoders. One forward pass over the encoder serves all three triage tasks, whereas three single-task systems pay the encoder cost three times per tweet. Inference latency and memory therefore become measurable benchmark axes in their own right. The MTO landscape is also consolidated enough to compare: LibMTL packages 13 optimization strategies and eight architectures behind one reproducible PyTorch interface (Lin & Zhang, 2023).

Single reported scores carry a variance risk that single runs hide. Reimers and Gurevych (2017) retrained sequence-tagging networks with identical hyperparameters but different random seeds. Maximum score differences reached 2.59 F1 points on the Conference on Computational Natural Language Learning (CoNLL) 2003 named entity recognition benchmark. The gap reached 8.23 F1 points on the Automatic Content Extraction (ACE) 2005 Entities task. Selecting a run by its development score was also unreliable, with Spearman’s rank correlation between development and test scores at only $\rho = .23$ in the reported case. The proposed benchmark therefore trains every

configuration under multiple seeds and treats single scores as insufficient evidence.

Comparison across many systems and datasets adds its own statistical constraints. Demšar (2006) showed that parametric tests rest on assumptions that comparisons across multiple datasets violate. The recommended alternative is the Friedman test with corresponding post hoc procedures, using the Nemenyi procedure for all-pairwise comparisons, together with the Iman–Davenport correction for the conservative Friedman statistic. The proposed benchmark adopts this nonparametric protocol for its method comparisons.

The last standard concerns the study’s form. Peffers et al. (2007) formalized design science research methodology as a process model of six activities in a nominal sequence, a framework for research that creates and evaluates artifacts. The proposed study follows that framing: The benchmark corpus and the prototype triage service are its design artifacts. No reviewed crisis study combines latency and memory measurement, multi-seed evaluation, nonparametric comparison, and artifact framing in one benchmark, which is the standards gap that Section 2.5 assembles.

2.2 Multitask Optimization Dynamics

Training one network on several tasks turns optimization into a balancing problem. Shared parameters serve every task loss at once, so an update that helps one task can hurt another. Two algorithmic answers dominate the response literature. Gradient surgery edits the shared update direction, whereas dynamic loss weighting adapts how much each loss counts. A third response acts on the architecture, using attention modules that modulate the shared representation across tasks (S. Liu et al., 2019), and the proposed benchmark

holds that side fixed. Sener and Koltun (2018) supplied the formal frame, casting multitask learning as a search for Pareto-optimal solutions, and their MGDA-UB (upper-bound objective) approximation anchored the optimization line this section reviews.

The surgery family begins with projecting conflicting gradients (PCGrad), whose authors named the obstacle precisely. Yu et al. (2020) defined two gradients as conflicting when they point away from each other, that is, when their cosine similarity is negative. Their repair projected a task's gradient onto the normal plane of any conflicting gradient before the update, a general procedure they called "gradient surgery," whose conflicting-gradient instance is PCGrad. On a series of challenging multitask supervised and multitask reinforcement learning problems, the approach yielded substantial gains in efficiency and performance.

Successors then reshaped the surgery idea along three main lines, with two additional context methods. B. Liu et al. (2021) introduced Conflict-Averse Gradient Descent (CAGrad), which maximized worst-case improvement near the average gradient while minimizing average loss, with a convergence guarantee earlier projection methods lacked. L. Liu et al. (2021) contributed Impartial Multi-Task Learning (IMTL-G), which rescaled per-task gradients in closed form so that the aggregated gradient carried equal projections onto every task. Nash multitask learning (Nash-MTL) replaced pairwise repair with a principled global rule, adopting the Nash bargaining solution from cooperative game theory (Navon et al., 2022). Two context methods completed the family. Independent Component Alignment for Multi-Task Learning (Aligned-MTL) aligned principal components via singular value decomposition (Senushkin et al., 2023). Similarity-aware momentum gradient surgery (SAM-GS) equalized gradients when

magnitude similarity signaled conflict and modulated momentum otherwise ([Borsani et al., 2025](#)).

The weighting family tunes scalar loss weights rather than editing gradient directions directly. Kendall et al. ([2018](#)) balanced loss functions by weighting each task with its homoscedastic (task-dependent) uncertainty, which lets noisy tasks count less. Gradient Normalization (GradNorm) learned the weights by steering per-task gradient norms toward targets set by relative inverse training rates ([Chen et al., 2018](#)). Fast Adaptive Multitask Optimization (FAMO) reduced the memory and time cost of gradient-manipulation methods by equalizing relative loss-improvement rates from loss history without storing per-task gradients ([B. Liu et al., 2023](#)).

Controlled comparisons then tested whether specialized balancing improves results. Kurin et al. ([2022](#)) defended unitary scalarization, training that sums the task losses. Coupled with single-task regularization and stabilization, it matched or improved upon complex multitask optimizers in popular supervised and reinforcement learning settings. Most of those optimizers require per-task gradients and introduce memory, runtime, and implementation overhead that scales with task count. Xin et al. ([2022](#)) reached the same verdict at scale, finding no improvements across language and vision tasks beyond what traditional optimization approaches achieve. Elich et al. ([2024](#)) questioned the mechanism itself, finding no evidence that angular gradient conflict is unique to multitask learning and locating the distinguishing factor in gradient magnitude. Gama and Grassi ([2025](#)) added the boundary condition: Specialized optimizers are unnecessary on simple problems, and their value re-emerges as task counts and feature-space demands grow.

One optimizer confound affects the multitask comparison dispute. Researchers

have often paired balancing rules with different optimizers, so a reported gain may reflect the optimizer rather than the balancing rule. Loshchilov and Hutter (2019) introduced AdamW, decoupling weight decay from the gradient-based update. Elich et al. (2024) derived a partial loss-scale invariance under mild assumptions, showing that loss scaling cancels out in head parameters while still affecting the shared backbone. The procedural remedy is direct: match budgets, hold the optimizer fixed, and tune baselines as carefully as the proposed methods (Elich et al., 2024; Kurin et al., 2022; Xin et al., 2022).

The claims of the surgery and weighting families and the counterclaims from the controlled comparisons therefore remain unresolved in the general domain apart from the complexity boundary condition. Researchers on neither side have tested that dispute on code-switched, low-resource text, and no reviewed study in that literature has compared optimization strategies under controlled conditions. That untested intersection is the evidence gap, and Section 2.3 examines the code-switched literature behind that gap.

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